

Example 8.1

$$X_t - \mu = \phi_1 (X_{t-1} - \mu) + Z_t$$

Parameters: μ, ϕ_1, σ_z^2

Using the Yule-Walker eq'ns

$$\hat{\Phi}_p = R_p^{-1} r_p \quad \leftarrow \text{vector of sample autocorrelations}$$

Here $R_1 = [1]$

$$\therefore \hat{\Phi}_1 = 1 \times r_1 \\ = r_1$$

$$\begin{pmatrix} r_1 \\ r_2 \\ \vdots \\ r_p \end{pmatrix}$$

$$R_p = \begin{bmatrix} 1 & r_1 & \dots & r_{p-1} \\ r_1 & 1 & & \\ & & \ddots & \\ & & & 1 \end{bmatrix}$$

$$\hat{\sigma}_z^2 = c_0 (1 - \hat{\Phi}_1 r_1) \\ = c_0 (1 - r_1^2)$$

$$c_0 = \frac{1}{n} \sum_{t=1}^n (x_t - \bar{x})^2$$

Finally $\hat{\mu} = \frac{1}{n} \sum_{t=1}^n x_t = \bar{x}$